

Origin, Dilemma and Development of the DFP Algorithm

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Outline

- 1 Introduction to the DFP Algorithm
- 2 Superlinear Convergence of the DFP Algorithm
- 3 Global Convergence of the DFP Algorithm
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unconstrained optimization

$$\min_{x \in \mathbb{R}^n} f(x), \quad f \in \mathcal{C}^2(\mathbb{R}^n)$$

Newton's method:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$$

- **Pros:** Quadratic convergence near the solution.
- **Cons:** Computing $\nabla^2 f(x_k)$ costs $O(n^2)$; inversion $O(n^3)$.

Quasi-Newton idea: Replace Hessian by an approximation B_k built *only from gradient information*.

The First Quasi-Newton Method: DFP

- **1959:** Davidon¹ proposes the first quasi-Newton algorithm (Argonne National Laboratory technical report).
- **1963:** Fletcher & Powell² systematically revise and popularize it.

The resulting DFP update introduces the secant condition:

$$B_{k+1}s_k = y_k, \quad s_k = x_{k+1} - x_k, \quad y_k = g_{k+1} - g_k$$

Where B_k approximates the Hessian and $g_k = \nabla f(x_k)$.

¹Davidon, W. C., *Variable metric method for minimization*, SIAM J. Optim., 1(1):1–17, 1991.

²Fletcher, R., Powell, M. J. D., *A rapidly convergent descent method for minimization*, Comput. J., 6(2):163–168, 1963.

Wolfe Conditions

Before discussing the specific algorithm, we first introduce the Wolfe conditions for step length selection.

Weak Wolfe conditions:

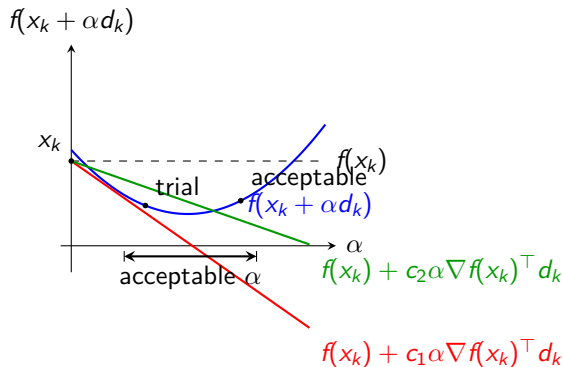
$$f(x_k + \alpha d_k) \leq f(x_k) + c_1 \alpha \nabla f(x_k)^\top d_k$$

$$\nabla f(x_k + \alpha d_k)^\top d_k \geq c_2 \nabla f(x_k)^\top d_k$$

where $0 < c_1 < c_2 < 1$ (e.g., $c_1 = 10^{-4}$, $c_2 = 0.9$).

Strong Wolfe conditions: replace curvature with

$$|\nabla f(x_k + \alpha d_k)^\top d_k| \leq -c_2 \nabla f(x_k)^\top d_k.$$



DFP update:

$$B_{k+1} = B_k - \frac{B_k s_k y_k^\top + y_k s_k^\top B_k}{y_k^\top s_k} + \left(1 + \frac{s_k^\top B_k s_k}{y_k^\top s_k}\right) \frac{y_k y_k^\top}{y_k^\top s_k}$$

Key property:

If B_k is symmetric positive definite and $y_k^\top s_k > 0$, then B_{k+1} is also symmetric positive definite.

- The condition $y_k^\top s_k > 0$ is guaranteed by the Wolfe line search (for convex f).
- Positive definiteness ensures $p_k = -B_k^{-1} g_k$ is a descent direction.

The BFGS Method

BFGS update (Broyden, Fletcher, Goldfarb, Shanno, 1970)³:

$$B_{k+1}^{\text{BFGS}} = B_k - \frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}$$

Let $H_k = B_k^{-1}$. Then the inverse update has the **same form**:

$$H_{k+1}^{\text{BFGS}} = H_k - \frac{H_k y_k y_k^\top H_k}{y_k^\top H_k y_k} + \frac{s_k s_k^\top}{y_k^\top s_k}$$

³Broyden, C. G., *The convergence of a class of double-rank minimization algorithms*, J. Inst. Math. Appl., 6:76–90, 1970.

The Broyden Family

DFP and BFGS belong to a one-parameter family:

$$B_{k+1}(\phi_k) = (1 - \phi_k) B_{k+1}^{\text{BFGS}} + \phi_k B_{k+1}^{\text{DFP}}, \quad \phi_k \in [0, 1]$$

or equivalently

$$B_{k+1} = B_k - \frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k} + \phi_k (s_k^\top B_k s_k) v_k v_k^\top,$$

where $v_k = \frac{y_k}{y_k^\top s_k} - \frac{B_k s_k}{s_k^\top B_k s_k}$.

- $\phi_k = 0$: BFGS
- $\phi_k = 1$: DFP

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Superlinear Convergence

Definition (Superlinear rate)

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0.$$

Theorem (Dennis & Moré, 1971)

⁴Let $x_{k+1} = x_k - B_k^{-1} \nabla f(x_k)$ converge to x^* with $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ nonsingular. Then convergence is superlinear **if and only if**

$$\lim_{k \rightarrow \infty} \frac{\|[B_k - \nabla^2 f(x^*)](x_{k+1} - x_k)\|}{\|x_{k+1} - x_k\|} = 0.$$

⁴Dennis, J. E., Moré, J. J., *A characterization of superlinear convergence and its application to quasi-Newton methods*, Math. Comp., 28:549–560, 1974.

Superlinear Convergence Results

- **Exact line search (Powell, 1971):**⁵

For uniformly convex f , DFP converges superlinearly.

- **Inexact line search (Wolfe conditions, Broyden family):**⁶

All methods in the Broyden family (including DFP) satisfy the Dennis condition and thus converge superlinearly, provided the iterates converge.

⁵Powell, M. J. D., *On the convergence of the variable metric algorithm*, J. Inst. Math. Appl., 7:21–36, 1971.

⁶Byrd, R. H., Nocedal, J., Yuan, Y., *Global convergence of a class of quasi-Newton methods on convex problems*, SIAM J. Numer. Anal., 24:1171–1190, 1987.

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The Course of Global Convergence Research

- **1971 (Powell)⁷**: DFP with *exact* line search converges globally for convex functions.
- **1976 (Powell)⁸**: BFGS with *inexact* (Wolfe) line search converges globally.
- **1987 (Byrd, Nocedal & Yuan)⁹**: All convex Broyden family methods converge globally except DFP.
- **1995 (Yuan)¹⁰**: Proposes sufficient conditions for DFP with weak Wolfe line search.
- **1995–present**: No definitive resolution for DFP with inexact line search.

⁷Powell, 1971.

⁸Powell, M. J. D., *Some global convergence properties of a variable metric algorithm for minimization without exact line searches*, in Nonlinear Programming, SIAM-AMS Proceedings, Vol. IX, pp. 53–72, 1976.

⁹Byrd, Nocedal, Yuan, 1987.

¹⁰Yuan, Y., *On the convergence of DFP algorithm*, Annals of Operations Research, 103:87–99, 2001 (Received 1995).

Zoutendijk Condition

Theorem (Zoutendijk, 1970)

¹¹Suppose:

- f is bounded below and ∇f is Lipschitz continuous.
- The line search satisfies the **Wolfe conditions**.
- p_k is a descent direction.

Then

$$\sum_{k=0}^{\infty} \cos^2 \theta_k \|\nabla f_k\|^2 < \infty,$$

$$\text{where } \cos \theta_k = \frac{-\nabla f_k^\top p_k}{\|\nabla f_k\| \|p_k\|}.$$

¹¹Zoutendijk, G., *Nonlinear programming, computational methods*, in Integer and Nonlinear Programming, J. Abadie, ed., North-Holland, 1970.

From Zoutendijk to Trace Control

For quasi-Newton methods:

$$p_k = -B_k^{-1} g_k, \quad \cos \theta_k = \frac{g_k^\top B_k^{-1} g_k}{\|g_k\| \|B_k^{-1} g_k\|}.$$

Step 1: Lower bound via eigenvalues.

$$\cos \theta_k \geq \frac{1}{\kappa(B_k)}, \quad \kappa(B_k) = \frac{\lambda_{\max}(B_k)}{\lambda_{\min}(B_k)}.$$

Step 2: Condition number bounded by traces. For a positive definite matrix:

$$\kappa(B_k) \leq \operatorname{tr}(B_k) \operatorname{tr}(B_k^{-1}).$$

(Since $\lambda_{\max} \leq \operatorname{tr}(B_k)$ and $1/\lambda_{\min} \leq \operatorname{tr}(B_k^{-1})$.)

Trace Control in the Broyden Family

From the Broyden family update¹²:

$$\text{tr}(B_{k+1}) = \text{tr}(B_k) + \frac{\|y_k\|^2}{y_k^\top s_k} + \phi_k \frac{\|y_k\|^2}{y_k^\top s_k} \frac{s_k^\top B_k s_k}{y_k^\top s_k} - (1 - \phi_k) \frac{\|B_k s_k\|^2}{s_k^\top B_k s_k} - 2\phi_k \frac{y_k^\top B_k s_k}{y_k^\top s_k}.$$

Crucial term:

$$\frac{\|B_k s_k\|^2}{s_k^\top B_k s_k} \geq \frac{\lambda_k}{c_2 \cos^2 \theta_k}.$$

And the step lengths satisfy:

$$\prod_{j=1}^k \lambda_j \geq c_4^k, \quad c_4 > 0.$$

From this, we can prove that all methods in the Broyden family except for DFP are globally convergent.

¹²R. H. Byrd, J. Nocedal, Y.-X. Yuan, 1987.

Assumptions:

- (i) f is twice continuously differentiable and **uniformly convex**:

$$\exists c_3 > 0, \quad p^\top \nabla^2 f(x) p \geq c_3 \|p\|^2, \quad \forall p, x.$$

- (ii) The line search satisfies the **weak Wolfe conditions**.

Under these assumptions, Yuan proved four **sufficient conditions** for global convergence of DFP¹³.

- $\alpha_k \geq \varepsilon > 0$ and the line search satisfies the **strong Wolfe conditions**.
- $\|g_{k+1}\|_2 \leq \|g_k\|_2$ for all sufficiently large k (monotone gradient norm).
- $\sum_{k=1}^{\infty} \|s_k\| < \infty$.

¹³Yuan, 1995.

Sufficient Condition 4

Theorem (Yuan, 1995)

Under uniform convexity and weak Wolfe line search, if

$$y_k^\top B_k g_k \leq 0 \quad \text{for all } k,$$

then the DFP algorithm converges globally.

Yuan proved the following inequality to control the trace of B_{k+1}^2 :

$$\text{tr}(B_{k+1}^2) \leq \text{tr}(B_k^2) + C \frac{1}{\|d_k\|^2}.$$

Then he used it to control $\text{tr}(B_k)$ and prove global convergence.

Limitations of Yuan's Conditions

- All four conditions are sufficient for global convergence, but they are difficult to check in practice within the standard DFP framework.
- Yuan also proposed two open inequalities:

$$\sum_{i=1}^k \frac{y_i^\top B_i s_i}{s_i^\top y_i} \leq c_{12} \sum_{i=1}^k \alpha_i,$$

$$\sum_{i=1}^k \left| \frac{y_i^\top B_i^2 s_i}{s_i^\top y_i} \right| \leq c_{13} \sum_{i=1}^k \frac{1}{\|d_i\|^2},$$

neither has been proven or disproven.

- This reflects that the DFP algorithm's convergence behavior is not ideal – in particular, the step length $\|s_k\|$ is not well controlled.

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Projection DFP

Therefore, Yuan, Zhou and Pham¹ modify DFP by projecting the step at out-points to enforce $\sum \|s_k\| < \infty$.

Define a constant $\tau > 0$ (e.g., $\tau = 0.01$) and the set of **in-points**:

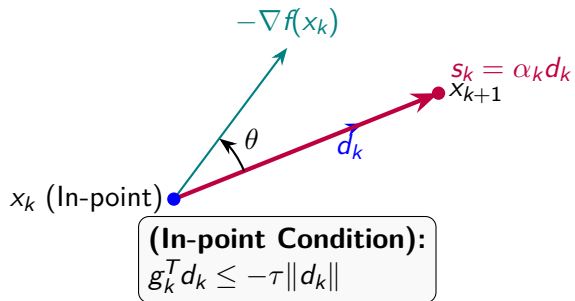
$$\mathcal{PX}_k = \{x_k \mid g_k^\top d_k \leq -\tau \|d_k\|\}.$$

Points not in $\mathcal{PX}_k$ are **out-points**.

For in-points, the iteration points are the same as standard DFP:

$$x_{k+1} = x_k + \alpha_k d_k.$$

¹Yuan, G., Zhou, P., Pham, H. T., *A projection DFP quasi-Newton algorithm and its applications*, Numerical Algorithms, 2026.



Projection DFP

For an out-point ($x_k \notin \mathcal{PX}_k$), let $\tilde{h}_k = x_k + \alpha_k d_k$.

Define the **parabolic surface**:

$$\{x \mid g(\tilde{h}_k)^\top d_k - \sigma_1 g(x)^\top d_k + \sigma_2 \tau \|d_k\| = 0\},$$

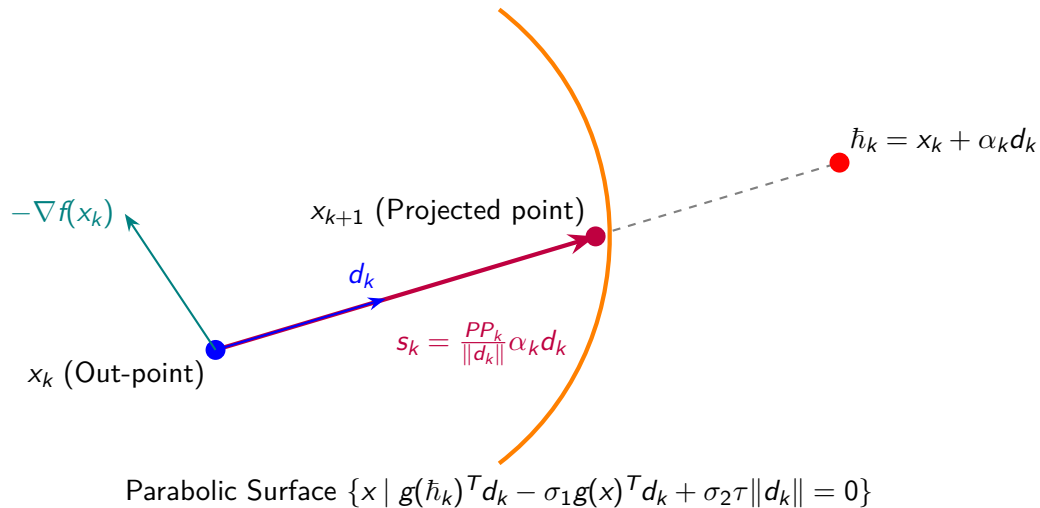
where $\sigma_1 + \sigma_2 < 1$ with $\sigma_1 \in (0, c_2)$ and $\sigma_2 \in (c_2 - \sigma_1, 1)$ are constants, and c_2 is the parameter in the weak Wolfe conditions. (e.g., $\sigma_1 = 0.425$, $\sigma_2 = 0.5$, $c_2 = 0.85$)

The projected iterate is

$$x_{k+1} = x_k + \frac{PP_k}{\|d_k\|} \alpha_k d_k, \quad PP_k = g(\tilde{h}_k)^\top d_k - \sigma_1 g(x_k)^\top d_k + \sigma_2 \tau \|d_k\|.$$

Then x_{k+1} lies on the parabolic surface, and the step is shortened.

Projection DFP



Standard DFP: $s_k = \alpha_k d_k$ always. If $g_k^\top d_k$ is very small, $\|s_k\|$ may grow very quickly.

Projection DFP:

- In-point: $g_k^\top d_k \leq -\tau \|d_k\| \Rightarrow \|s_k\| \leq \frac{1}{\tau} (-g_k^\top s_k)$.
- Out-point: Projection shortens the step: $\|s_k\| = \frac{PP_k}{\|d_k\|} \alpha_k \|d_k\|$ with $\frac{PP_k}{\|d_k\|} < 1$. Then we can prove that $PP_k \leq -\sigma_3 g_k^\top d_k$. Hence $\|s_k\| \leq -\sigma_3 \alpha_k g_k^\top d_k$.

Global Convergence of Projection DFP

So we can know that:

$$\|s_k\| \leq C \cdot (-\alpha_k g_k^\top d_k) \quad \text{for all } k,$$

where $C = \max\{1/\tau, \sigma_3\}$.

By Wolfe conditions, we can prove that

$$\sum_{k=0}^{\infty} (-\alpha_k g_k^\top d_k) < \infty.$$

Therefore

$$\sum_{k=0}^{\infty} \|s_k\| < \infty.$$

Hence the algorithm obviously converges globally.

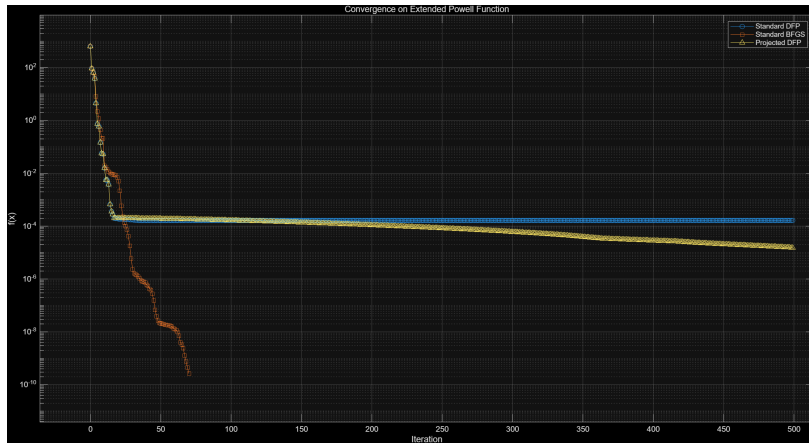


Figure: Comparison experiment

References



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G. Yuan, P. Zhou, H. T. Pham, *A projection DFP quasi-Newton algorithm and its applications*, Numerical Algorithms, 2026.

Thank you!

Thanks for your attention!

Questions?